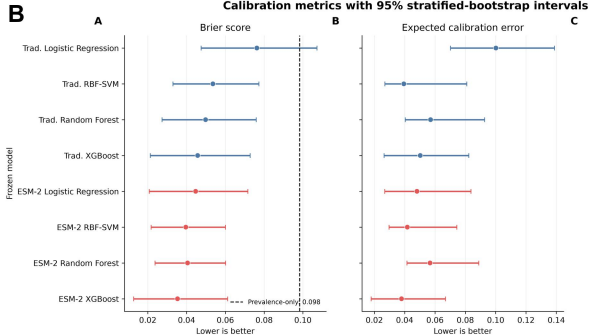
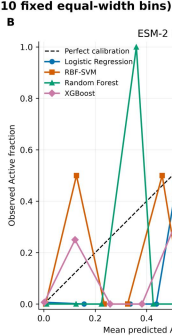
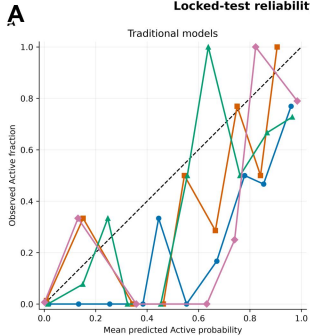
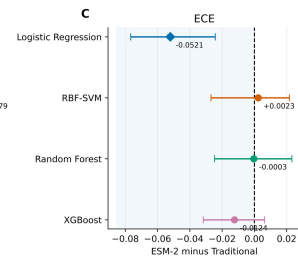
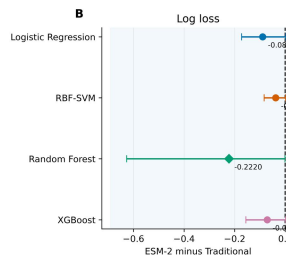
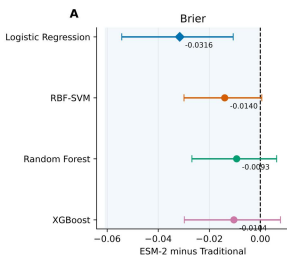


**A Locked-test reliability curves (10 fixed equal-width bins)**

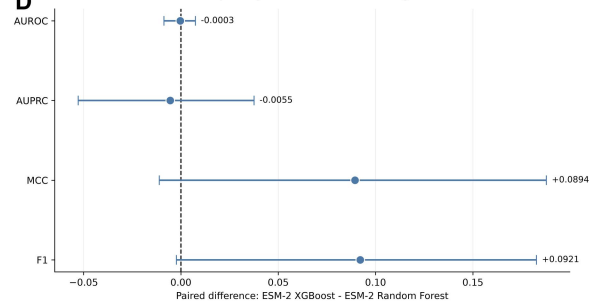


**Paired calibration differences: ESM-2 vs traditional descriptors**



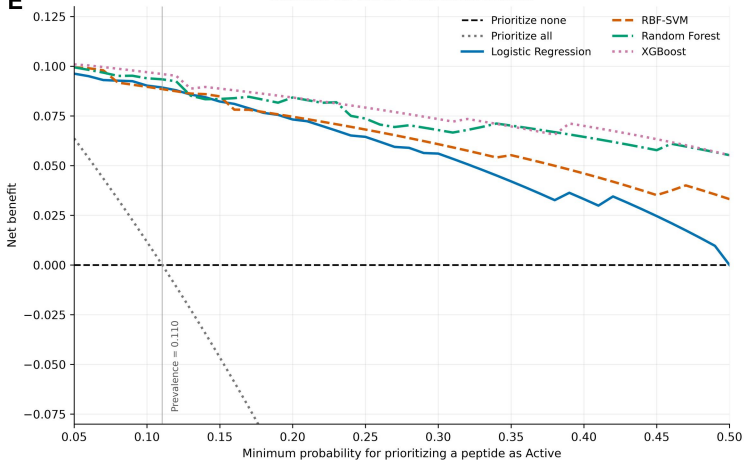
→ Negative values favor ESM-2 | Positive values favor traditional descriptors →  
Diamonds: 95% paired percentile interval excludes zero; circles: interval includes zero. Descriptive, not multiplicity-adjusted.

**Paired Bootstrap Comparison of the Two Leading ESM-2 Models**



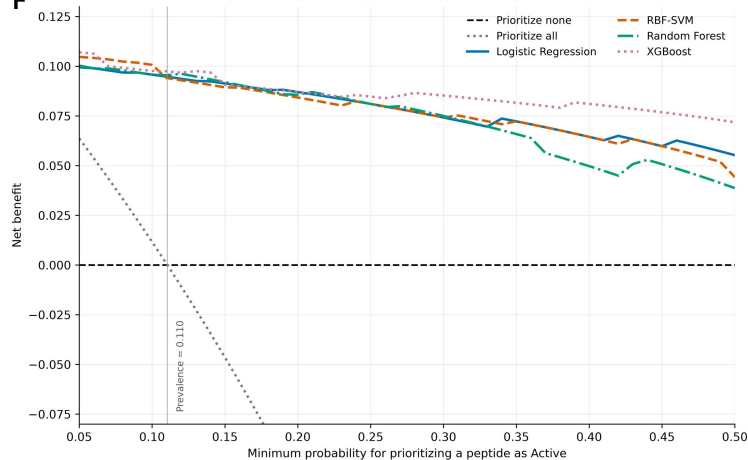
Positive values favor XGBoost; negative values favor Random Forest.  
Diamonds: 95% paired percentile interval excludes zero; circles: interval includes zero. No multiplicity-adjusted significance claim is made.

**Decision curves for Traditional models**



Higher net benefit is preferable. Thresholds were predefined and were not used to retune, recalibrate, or select a model.

**Decision curves for ESM-2 models**



Higher net benefit is preferable. Thresholds were predefined and were not used to retune, recalibrate, or select a model.